

Can the Minutes Pipeline Go Region-Wide?

The Bay Area reconnaissance sprint, June 2026 — consolidated findings from nine probes

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August 2026

Standing document. The single memo for `output/bay_area_recon/`. It replaces the nine per-probe memos that used to sit one per directory; each probe directory keeps its CSVs, scripts and `.md` reports, and §6 says which file came from where. Where a narrative `.md` report disagrees with its own CSV, the CSV is authoritative and the disagreement is recorded in §7 rather than quietly resolved.

1 The question, the answer, and what is still blocked

The project studies the *market for housing regulation*: an item-level dataset of discretionary land-use decisions built from San Francisco Planning Commission minutes (1998–present), read through a model with two margins — the *by-right envelope* $\bar{z}_j$ a locality allows without a hearing, and the *discretionary* margin τ_j applied to everything else. The headline test is substitution: when the state forces the by-right margin open, do localities claw restriction back through discretion? That test needs neighbours, so this sprint asked one logistical question — **can the SF pipeline be extended to the roughly 109 land-use regulators of the nine-county ABAG Bay Area?** Nine probes ran in June 2026.

Yes, with one hard qualification and one unmade decision. Retrieval is tractable: all 109 localities were surveyed and keyed to a Census GEOID, the region’s meeting portals collapse to four vendor families covering 87 of the 109, and only **three** sites region-wide are genuinely bot-blocked. Depth is adequate for the window that matters: **36 localities** have minutes reaching 2016 or earlier, straddling the 2017–2023 preemption ramp, and 36 is a floor, not a ceiling. The spatial side is better — **67 of 109** publish a downloadable zoning GIS layer and a statewide layer backstops the rest, so address-to-district matching is a script, not an RA-hours problem. The treatment variable is clean: all 25 estimation-sample localities carry sourced builder’s-remedy exposure dates spanning **0.0 to 38.4 months**.

The qualification is extraction. The SF heuristics **do not transfer**: run unmodified on a real Daly City minutes document, `autoextract.extract()` produced two correct fields out of roughly 35, and three of the wrong ones were populated with plausible-looking garbage. Region-wide expansion is a research programme with a per-jurisdiction (or LLM) extractor and a human in the loop, not a scripting exercise.

The unmade decision gates the whole right-hand side: **does the by-right envelope mean “principally permitted use” or “no discretionary review of any kind”?** Fremont and San Mateo both apply design or site review to uses their codes mark permitted, so the code’s “P” column is strictly wider than the model’s ministerial envelope. Nothing downstream is well defined until this is settled, and no further data collection will settle it.

Blocked, in order of consequence. (i) The envelope definition above. (ii) **San Jose**, whose 2005–2018 minutes are confirmed to exist but sit behind Akamai — a budget call, not a measurement. (iii) The **41 localities with unknown archive depth**, of which 12 are CivicPlus sites needing one browser session to read a committee ID. (iv) **Sample width**: both the HCD panel and the pre-period probe were fixed at 25 localities, and depth work has since raised confirmed ≤ 2016 breadth to 36. (v) The **pre-period envelope for 10 of 25** sample localities, a per-publisher superseded-version dig rather than archival research.

2 The frame: 109 land-use regulators, keyed and classified

Everything joins to `bay_area_census/bay_area_locality_census.csv` — 109 rows $\times$ 20 columns, one per entity that actually regulates land use: 101 incorporated cities and towns, the eight county governments, and San Francisco as a consolidated reference row (85 typed `city`, 16 `town`, 8 `county`). The city frame was supplied rather than agent-generated, so completeness is not in question; eight per-county shallow surveys in `raw/` (Alameda 15, Contra Costa 20, Marin 12, Napa 6, San Mateo 21, Santa Clara 16, Solano 8, Sonoma 10 = 108 localities) are the provenance for every cell. Each row records the regulating body, minutes URL, vendor platform, code host, an access status per side, and — the point of the exercise — a *solution class*: the single engineering effort that would unlock it, which turns a 109-item to-do list into a handful of adapters.

The region collapses into a few adapters. Minutes side: `granicus_family_clean` 41, `civicplus_clean` 26, `custom_cms` 10, `civicclerk_modern_api` 10, `primegov_portal` 4, `civicweb_portal` 3, `unknown` 3, `civicplus_akamai` 2, `escribe_portal` 2, `onbase_portal` 2, `municode_meetings_portal` 2, and one each of `novusagenda_portal`, `laserfiche_portal`, `granicus_family_akamai` and `reference_done` (San Francisco). Four classes — Granicus-family, CivicPlus, CivicClerk and the custom-CMS tail — cover **87 of 109**, and a single Granicus/Legistar adapter unlocks 41 on its own: the highest-ROI engineering in the project. Zoning code hosts cluster harder: `municode_api` 44, `codepublishing_clean` 23, `ecode360_clean` 14, `publiclaw_clean` 8, `amlegal_or_ecode_403` 7, `unknown` 6, `city_site` 3, `qcode_clean` 2, and one each `encodeplus` and `reference_done` — three hosts cover 81 of 109.

Access is largely solved, and the Akamai scare was a myth. Measured `minutes_access`: `clean` 78, `js_shell` 16, `unknown` 10, `akamai_403` 3, `unsupported_platform` 2. The 16 `js_shell` rows are not blocked — their content sits behind a modern API (CivicClerk, PrimeGov, IQM2, OnBase, Municode Meetings) and needs an API client. Zoning access: `clean` 94, `unknown` 14, `akamai_403` 1. Tractability tier: `needs_adapter` 69, `clean` 35, `access_blocked` 3, `unknown` 2. The census was written on the premise that the roughly 27 CivicPlus localities might *all* sit behind Akamai — a class-wide wall. A header-only sweep of all **28** CivicPlus hosts with an ordinary browser User-Agent (no evasion, no header tricks, no IP rotation) found **25 clean**, **2 Akamai 403s** (Fremont, Portola Valley) and **1 stale-URL 404** (Antioch, behind Cloudflare — a wrong path, not a bot wall). Akamai is a per-site subscription, not a platform property; counting San Jose, the region has **three** genuinely blocked sites. The working retrieval path was demonstrated end-to-end: `fetch /AgendaCenter` with a normal User-Agent, parse the embedded `ViewFile/Minutes/...` links, walk `/AgendaCenter/PreviousVersions` for history. The 19 KB Saratoga minutes PDF in `bay_area_census/` proves the path yields a real document.

The join key is solid; the body names are not. All 109 rows carry a real GEOID and zero are `to_verify`; names were never the join, because name joins fail silently here (NZLUD stores Redwood City as “Redwood”). Row confidence: `med` 64, `high` 41, `low` 4. Minutes are recorded as posted for 104 localities, `unknown` for 4, `sparse` for 1; none was confirmed paper-only, so there is no digitization emergency, and the five whose platform could not be identified are Colma, Monte Sereno, Newark, Fairfax and San Rafael. The regulating body is often *not* a “Planning Commission” — Planning Board (Alameda), Planning & Zoning Commission (Albany), Environmental Planning Commission (Mountain View), Planning & Transportation Commission (Palo Alto, San Carlos), a combined Planning Commission and Board of Zoning Adjustments (San Leandro), Permit Sonoma — so cross-jurisdiction matching on body name will break.

Left open here. Ten non-CivicPlus `minutes_access = unknown` rows were never probed; six zoning hosts and three minutes platforms remain `unknown`; eight localities run dual or migrating systems where the authoritative record is not established (Foster City, San Pablo, Menlo Park, El Cerrito, Pinole, Union City, Brentwood, Santa Clara County); Antioch’s correct agenda URL is still wrong in the CSV; and a per-host terms-of-service check before any bulk run remains prudent.

3 Minutes: how far back, and can we fetch them

Four probes answer one question. `archive_depth_probe/archive_depth.csv` (109×10) is the post-merge authority on depth; `civicplus_depth_probe/civicplus_depth.csv` and `migration_cliffs_probe/mi` feed corrections into it; `minutes_platform_pilot/` is the only probe that touched an actual document.

3.1 Depth distribution and the panel window

Pass 1 read each of the 109 archives as a normal browser client — no downloads, time-boxed, *unknown over guess* — and left 55 of 109 unknown, mostly because JS-shell portals do not put depth in static HTML. **Pass 2** (`api_probe.py`, 22 live jobs) queried each portal’s own API for metadata only: Legistar `events` filtered on `EventMinutesFile ne null`, CivicClerk `/v1/Events` scanning `publishedFiles` for `type="Minutes"`, PrimeGov `ListArchivedMeetings`, and the Granicus minutes-mode RSS feed (capped at roughly 80–100 items). **The typing rule** is what makes the numbers mean anything: a year counts only if the API returned an actual Minutes-type file — agendas, packets, video and year-dropdown ranges do not count.

Earliest minutes year	Count	Share	Common start	Localities	high/med
≤ 2005	6	5.5%	≤ 2014	29	22
2006–2010	11	10.1%	≤ 2016	36	29
2011–2015	15	13.8%	≤ 2018	45	37
2016 and later	36	33.0%			
unknown	41	37.6%			
Known depth	68	62.4%			

A 2016-start panel therefore has enough co-submarket breadth for a strategic-interaction reaction function around the ramp, and 36 is a **floor**: the 41 `unknown` rows are overwhelmingly a method limit (JS or API-gated portals), not proven-shallow archives, so resolving them can only add localities. Deep history is the opposite story — only 6 localities reach 2005 or earlier and

only 17 reach 2010 or earlier — so long-run “ratchet” dynamics are not supported by the online record. Supporting distributions: confidence **high** 32, **med** 23, **low** 54; continuity **continuous** 48, **gaps** 3 (Cloverdale 2012 missing, Fairfax 2025 missing, Daly City sparse), **unknown** 58; type basis **typed_minutes** 42, **type_ambiguous** 30, **agendas_only** 2, **unknown** 35.

Of the 55 rows in `archive_depth_api.csv` (55×11), `minutes_type_confirmed` is **yes** for 20, **no_api** for 34, **absent** for 1. Nineteen resolved to a specific verified year: Alameda 2005; El Cerrito and San Carlos 2008; Albany and Daly City 2009; Oakland and Burlingame 2014; Hayward, Emeryville and Sonoma County 2015; Santa Rosa 2016; Hercules, Napa and San Mateo County 2017; Pinole and San Mateo 2019; Napa County 2021; Contra Costa County and Pleasanton 2023. The twentieth “yes” is the city of Sonoma, whose CivicWeb portal confirms a Minutes bucket labelled “prior to 2017” without exposing the year. The one **absent** is **Cupertino**: Legistar holds events with status Final back to 2005 but `EventMinutesFile` is null throughout, so its minutes live elsewhere. Platform mix of the 55: Legistar/Granicus 20, CivicPlus 14, CivicClerk 5, custom CMS 4, PrimeGov 3, CivicWeb 3, Municode Meetings 2, and one each of eScribe, OnBase, NovusAgenda and other.

3.2 The agenda trap

Year-dropdown floors systematically overstate depth, because dropdowns span *agendas* or events, which predate posted minutes. Hard checks against Legistar `EventMinutesFile`:

Locality	Dropdown floor	Actual minutes	First minutes record
Oakland	2000	2014	2014-09-18
Santa Rosa	1999	2016	2016-02-02
Hayward	2013	2015	—
Alameda	2005	2005 (confirmed)	2005-01-12
Emeryville	2015	2015 (confirmed)	—

Oakland alone was a 14-year overstatement. Any remaining non-API dropdown floor among the 68 known rows is an *upper bound* until API-confirmed.

3.3 CivicPlus: the method works, the coverage does not

CivicPlus AgendaCenter exposes no depth API, leaving 14 localities as the largest unresolved block — but it does have a server-rendered Search endpoint answering a date-range query in plain HTML (`/AgendaCenter/Search/?term=&CIDs={cid}&startDate=...&endDate=...`) and returning links of the form `ViewFile/Minutes/YYYYMMDD-N`. Because the document type is *in the path*, the read is definitively minutes-typed — the property dropdown reads lacked, and the reason the agenda trap cannot bite here. Regex the planning body’s numeric CID out of `/AgendaCenter`, query Search, take the minimum year in `ViewFile/Minutes` links, collecting agenda years separately so an agenda/minutes gap is flagged rather than mistaken for depth. The first pass used one wide 2004–2018 window; the second walked single probe years (2008, 2010, 2012, 2014, 2016), avoiding truncation and giving a crude continuity check.

Result: 2 of 14 resolved. **Campbell 2006** is the win — minutes-typed files continuously 2006–2016 under `CID=6`, with 2004 empty: a clean verified floor, and one of only six Bay Area localities reaching 2006 or earlier. **Los Altos Hills 2022** is a floor, not a depth — static `ViewFile/Minutes` links for 2022–2026 with the deeper archive still gated, recorded at **med** confidence flagged `static.html_floor(deeper_may_exist)`. The other twelve — Antioch, Atherton, Cotati, Half Moon Bay, Millbrae, Monte Sereno, Oakley, Pleasant Hill, San Anselmo, San

Pablo, Windsor, Yountville — run the newer **CivicEngage** generation, which renders the committee list client-side: the CID never appears in static HTML and **PreviousVersions** returns 404, so the query cannot be formed. That is a *rendering* limit, not an access one (HTTP 200 throughout, hosts confirmed unblocked by the Akamai sweep), so all twelve are honest **unknown** flagged `civicengage.js_gated(CID_not_static)` — explicitly **not minutes_absent**. One browser-rendered session reading each CID out of the rendered DOM finishes the job; it is cheap and not a blocker, since the twelve are mostly small cities outside the estimation panel.

3.4 Migration cliffs: a change of address, not a loss of data

Three jurisdictions showed minutes apparently starting around 2024 because they migrated their portal that year — and they are among the largest regulators in the set (San Jose is the region’s biggest city; the two counties regulate all unincorporated territory), so a 2024 start would have excluded them from the ramp panel. Each prior platform was identified from redirects, clerk pages and URL conventions, tested for whether the legacy host still serves, and read at listing level using minutes-only feeds (Granicus `ViewPublisherRSS.php?mode=minutes`) or year-filtered calendars (IQM2). **All three cliffs were domain or view migrations. 3 of 3 gain deep history.**

Marin County — 2005, high confidence: Board of Supervisors agendas, minutes and video have been online since 2005 on the legacy Granicus view (`view_id=33`), and a separate older archive at `marincounty.org/depts/bs/oldmeetingarchive` carries 1995–2005 agendas and minutes without video. **Santa Clara County — 2008, medium:** the legacy IQM2 instance at `sccgov.iqm2.com/Citizens/` is still up and returns Board Minutes links back to 2008, with the CSV explicit that the type is strongly indicated but *not* document-verified and that the archive likely extends earlier. **San Jose — 2005 at the listing, documents gated:** the legacy Granicus minutes RSS (`view_id=51`) lists **80 minutes items spanning 2005–2018**, but the MinutesViewer link 302-redirects to the `sanjoseca.gov` DocumentCenter, which is Akamai-blocked; recorded as `prior_archive_reachable = listing_only` with an explicit instruction not to attempt to defeat the block.

All three merged into `archive_depth.csv`; together with Campbell 2006 and Los Altos Hills 2022 they took the confirmed ≤ 2016 panel **from 32 to 36 localities**. The lesson generalises — a portal migration is a *visible-archive cliff*, and any locality whose depth begins at a suspiciously round recent year should be checked against its prior platform — but only these three were investigated, so other cliffs may sit undetected inside the 41 **unknown** rows. Both surviving legacy archives are decommissioning candidates, so retrieving Marin and Santa Clara County history is time-sensitive.

3.5 The retrieval and extraction pilot: the one probe that touched a document

The earlier 14-city pilot (Daly City, South San Francisco, San Mateo, Redwood City, San Bruno, Burlingame; San Jose, Palo Alto, Mountain View, Sunnyvale; Oakland, Berkeley, Fremont, Richmond) is superseded in *coverage* by the census but is the only source for the sprint’s most consequential finding.

No single-vendor scraper covers the set. The 14 cities span six stacks: Legistar/Granicus $\times 6$, CivicPlus $\times 4$, CivicClerk $\times 1$, PrimeGov $\times 1$, Hyland OnBase $\times 1$, eScribe $\times 1$, plus Berkeley on a custom CMS with Laserfiche. Berkeley and the City of San Mateo could not be cleanly classified, and two platforms present — eScribe and Hyland OnBase — are unsupported by `civic-scraper` in any form.

One of three pilot cities yielded minutes. *Daly City* succeeded, but not through `civic-scraper`, which targets the legacy CivicClerk ASPX portal while *Daly City* runs the modern `*.api.civicclerk.com` React backend. The modern OData API does work: an `Events` query filtered on `categoryName eq 'Planning Commission'` returns events whose `publishedFiles[]` carries typed Agenda, Minutes and Agenda Packet entries, downloadable via `GetMeetingFileStream` — the pattern later reused for the roughly ten CivicClerk localities region-wide. *San Jose* failed on every route: its Legistar Web API returned complete metadata (14 Planning Commission meetings for the first half of 2024, matching the second-and-fourth-Wednesday schedule) but every one has `EventMinutesFile = False` and status “Draft”, and the Granicus route dead-ends at the Akamai-blocked Document-Center. *Fremont* failed on access: `civic-scraper` returned **zero assets**, and `/AgendaCenter`, `/government/agenda-center` and the RSS feed all return HTTP 403.

A silent-gap risk: publishing minutes is not the same as holding meetings. Of the 12 *Daly City* Planning Commission meetings in 2024, the API exposed Agenda $\times 11$, Agenda Packet $\times 5$, Minutes $\times 1$ — *one* minutes document for the year, so an unattended backfill would report success and return essentially nothing. Alongside it, minutes are approved one meeting later and attached to the *following* meeting’s packet (the 5 December 2023 minutes arrived under the 6 February 2024 packet), so a backfill must key minutes to meeting $N - 1$. The same pattern recurs in the census, where Campbell and Healdsburg show agendas but no posted minutes.

The central finding: the SF extractor does not transfer. Run unmodified on the real *Daly City* minutes, `autoextract.extract()` produced, out of roughly 35 schema fields, **two** correct ones — `ceqa_determination` (eir; an FEIR was before the commission) and `item` (1). The rest were empty or confidently wrong:

Field	Extractor	Truth	Failure mode
<code>case_number</code>	21-14998	GPA-04-21-14998	SF regex grabbed the tail of the case number
<code>units_proposed</code>	235	1,235	the thousands comma split the match
<code>parking_spaces</code>	5	34.5	the decimal point split the match
<code>project_address</code>	333 90Th Street	city-hall boilerplate	grabbed the page footer
<code>request_type</code>	empty	GPA + PD + subdivision + design review	SF uses case-number <i>suffixes</i> ; <i>Daly City</i> uses <i>prefixes</i>
<code>action</code>	empty	“voted 4-0 ... recommend approval”	no SF-style ACTION: label in the document
<code>assessor_block</code>	empty	APN 091-211-230	<i>Daly City</i> writes APN, not “Assessor’s Block”
<code>vote</code> , <code>speakers</code> , <code>stance</code>	empty / 0	4-0 vote; no stance markers	SF’s post-2015 stance markers do not exist here

The dangerous rows are the middle three: an empty field announces itself; a populated address, unit count and parking count that are all silently wrong do not. The operational conclusion, carried forward as a *gating quality step*: the SF heuristics must never run unattended on another jurisdiction, and any multi-jurisdiction build must first validate an extractor on real Granicus or CivicClerk minutes *with human review*.

Vocabulary does not transfer either. The proposed mapping layer (`jurisdiction_mappings.py`, additive, not wired into the pipeline) covers Daly City only: of its 13 `request_type` entries, five are clean synonyms (general plan amendment, GPA, rezoning, conditional use, variance) and eight are flagged judgment calls; of seven `action` entries, five are clean and two flagged. Three flagged cases matter substantively. **“DR” means different things in different cities:** in SF it is *Discretionary Review* — review of a project that is otherwise by right, precisely the behaviour the model is about — while in Daly City it is *Design Review*, so mapping on the letters would be exactly wrong. **Four common suburban request types have no canonical enum** — Planned Development, Major Subdivision, Precise Plan, Development Agreement — and all collapse to `other`; they recur across California suburbs, so it is a schema question. And **“recommend approval” is not approval:** on legislative items the Daly City commission only recommends to the City Council, while the SF `action` enums assume a final disposition. The pilot document is exactly that case — a 4-0 vote to *recommend* approval of a general plan amendment, rezoning, subdivision, design review and EIR certification for a **1,235-unit** project.

Tooling limits, recorded because they will recur. `civic-scraper` v1.1.0 has four defects that bit here: its CLI auto-routes only CivicPlus and DigitalTowPath; `LegistarSite` crashes in `pytz.timezone(None)` unless a timezone is passed; its Legistar events call fetches the current year only (`events(since=2024)` returned 180 events all dated 2026), so backfill needs the Legistar Web API with date filters; and its CivicClerk and CivicPlus scrapers target older portal versions than several Bay Area cities run.

4 Zoning: the current envelope and the pre-period envelope

4.1 Spatial form: a script, not an army of RAs

`zoning_map_form_probe/zoning_map_form.csv` (109×8) classifies the *form* in which each locality publishes its current spatial zoning — deliberately not what the zoning says — looking in order for a downloadable GIS layer, a viewer with no obvious download, a static PDF map, or nothing.

best_form	Count	Share	Meaning
<code>gis_layer</code>	67	61.5%	downloadable shapefile / GeoJSON / feature service
<code>gis_viewer_only</code>	28	25.7%	interactive viewer, no obvious download
<code>pdf_map</code>	13	11.9%	static zoning-map PDF only
<code>none_found</code>	1	0.9%	nothing located (Pacifica)

`download_apparent`: yes 71, no 37, unknown 1. Confidence: high 60, med 46, low 3 (Hillsborough, San Bruno, Pacifica). The 13 PDF-only localities — Atherton, Clayton, Colma, Daly City, Healdsburg, Hercules, Millbrae, Monte Sereno, Pinole, Portola Valley, San Pablo, Saratoga, Woodside — are all small, none a large regulator. Within `gis_layer`, **55 are native** city, county or consortium sources and **12 are covered only by the statewide fallback**: twelve Alameda County cities (Alameda, Albany, Dublin, Emeryville, Fremont, Hayward, Livermore, Newark, Piedmont, Pleasanton, San Leandro, Union City) whose own portals were never separately checked — verifiable from the file, since exactly 12 `gis_layer` rows carry a `gis.data.ca.gov` URL. Three regional or county sources close 26 localities on their own: MarinMap (all 12 Marin localities, 11 rows via `gis.marinpublic.com` plus the county’s own), the Solano Regional GIS Consortium (all 8 Solano localities on one ArcGIS organisation), and Napa County GIS (all 6 Napa localities, 5 on

gis.napacounty.gov plus the city of Napa’s own). Contra Costa, San Mateo, Santa Clara and Sonoma county GIS cover unincorporated areas only; their cities largely self-publish. Above it all sits the CA Statewide Zoning layer (Gov-OPR, gis.data.ca.gov), one downloadable layer plus REST, reported as covering **535 of 539** California jurisdictions.

Spatial matching is therefore a **scriptable problem**: native layers where they exist, the statewide layer as gap-filler for the 28 viewer-only, 13 PDF-only and 1 none-found localities, so no locality is a dead end for automation — decisively better than the minutes side, where roughly 38% still have unverified depth. **The one substantive risk** is that the statewide layer is a **2022–23 aggregated snapshot** of varying per-city fidelity, underpinning automation for about **54 of 109** rows once the 12 unchecked Alameda cities are counted; since the by-right envelope is the load-bearing right-hand-side variable, it should be spot-checked against each city’s own viewer or PDF — accept it as the universal first cut, or budget native extraction for the roughly 42 localities that natively offer only a viewer or a PDF. Open, cheapest first: confirm feature-service endpoints behind the 28 viewer-only sites (the true gis.layer count is likely above 67; the exceptions are El Cerrito, Lafayette and Martinez on the non-ArcGIS “CommunityView” product); check the 12 Alameda cities natively; resolve the three low-confidence identifications. Caught in passing: the brief’s claim that “20 rows already mention GIS” in the census was a substring artefact — the “gis” inside Legistar — so all 109 were probed fresh. Noted but not acted on: Stanford EarthWorks holds a **historical 2014 version of the San Mateo County planning-zones layer**, a pointer for the deferred historical-envelope workstream.

4.2 What the code says: NZLUD, code hosts, and the extraction pilot

zoning_envelope_project/ worked the same 14-city set as the minutes pilot, asking whether a machine-readable by-right dataset exists, whether the envelope can be read out of municipal codes at acceptable cost, and whether the preemption shock can be dated: mostly no, yes but not mechanically, and yes cleanly.

NZLUD covers 4 of 14, and only pre-preemption. The National Zoning and Land Use Database (Mleczko and Desmond, 2023) is an NLP-derived reading of municipal codes collected 2019–2022, replicating the 2006 Wharton WRLURI frame plus 210 supplemental municipalities; the full release in `_source_data/nzlud_muni.csv` is 2,639 municipalities $\times$ 77 columns, and its key field `mf_per` is the proportion of residential districts (including mixed-use) permitting multifamily housing *by right* — close to the margin the model needs. Matching on Census place FIPS yields exactly four rows. The ten absent cities include most of the Peninsula and every East Bay city except Fremont — Daly City, South San Francisco, San Mateo, San Bruno, Palo Alto, Mountain View, Sunnyvale, Oakland, Berkeley, Richmond — a direct consequence of the WRLURI-2006 frame, which no amount of waiting will fix.

City	mf_per	zri	height_ft_median	parking_median	Snapshot
Burlingame	0.667	2.925	50	2	2021-02-19
Redwood City	0.600	2.533	50	1	2019-10-24
Fremont	0.467	2.537	30	1	2019-10-15
San Jose	0.389	3.318	45	1	2021-12-19

Every snapshot predates the 2023–2026 sixth-cycle Housing Element updates and predates or coincides with the SB-9, SB-35 and SB-423 rezonings, so NZLUD is best read as a *pre-preemption* envelope: a limitation for measuring today’s rules, an opportunity for a two-period design. The

one available spot-check is close but not identical — Fremont’s Title 18 codes multifamily as **P** in R-3 and R-G and prohibited in R-1 and R-2, a naive base-residential by-right share of $2/4 = 0.50$ against NZLUD’s 0.467, the gap being a denominator disagreement (NZLUD counts mixed-use districts as residential) rather than an obvious error in either. San Jose could not be checked at all because Municode blocked the read; its 0.389 is unverified, and its Title 20 uses an **S** = Special Use Permit symbol easy to misread as by-right.

Half the code hosts block bots. All 14 codes were located: Municode $\times 6$, eCode360 $\times 3$, public.law $\times 2$, QCode $\times 2$, American Legal $\times 1$, CodePublishing $\times 1$ (South San Francisco is ambiguous between QCode and eCode360 and is counted in both). Only **CodePublishing** and **public.law** returned real code text to a plain GET: Municode returns HTTP 200 with a roughly 6 KB Angular shell and zero code text (the ordinance loads from a separate JSON API), while American Legal, eCode360 and Berkeley’s host return 403 outright. This matches the NZLUD authors’ own note that they could not use web scraping and downloaded codes manually.

Two structural traps, both fatal to a naive parser. Fremont’s code is a consolidated use matrix defining its own symbols (**P** permitted, **C** conditional use permit, **Z** zoning-administrator permit, **--** prohibited) and parsed cleanly. San Mateo splits permitted uses (§27.22.010) from special uses (§27.22.020) into *separate sections per district*, which a generic table parser would miss entirely; worse, §27.22.055 says building height “shall not exceed the standards set forth on the Building Height Plan of the General Plan” — **the height limit is not in the code text at all**. Height cannot be text-scraped for such cities, which also makes NZLUD’s height fields unreliable wherever a code defers to a map. Both pilot cities carry unresolved overlay districts (Fremont: TOD, hillside, housing-element-sites; San Mateo: two-unit and mixed-use residential).

The WRLURI benchmark disagrees with NZLUD. `wrluri_crosscheck.csv` (14 rows $\times$ 8 columns) shows NZLUD covering 4 of 14, WRLURI 2006 covering the *same* 4 of 14 (confirming the inherited frame), WRLURI 2018 covering 1 of 14 usable (Sunnyvale, 0.63), and all three overlapping on **zero** cities. Over the four-city overlap the indices are rank-inverted: Spearman between WRLURI 2006 and NZLUD `zri` is -1.00 where the expected sign is positive, and between WRLURI 2006 and `mf_per` is $+0.40$ where the expected sign is negative, with San Jose driving it — least restrictive of the four by the 2006 survey, most restrictive by NZLUD’s code index. This is neither validation nor refutation: $n = 4$ makes a ± 1.00 correlation nearly meaningless, the vintages are 13–15 years apart across California’s entire preemption shift, and the constructs differ by design. The takeaway is a caution — per-cell measures such as `mf_per` and height are more defensible to carry forward than either composite index.

4.3 The pre-period envelope: 15 of 25 recoverable

The difference-in-differences needs a *before* picture of the envelope, but codes and zoning maps are published as *current* documents, silently overwritten. The Wayback probe asks only whether a circa-2016 envelope can be *located and dated* cheaply across the 25-locality sample. In-window means a Wayback CDX capture timestamped **2014–2018**, and the capture timestamp *is* the vintage evidence — the probe never infers a date from content. Two forms count: an archived municipal-code page (text form, whose use table is the envelope) and an archived zoning-map PDF (spatial form). Deep code URLs were broadened to the publisher’s code-library root so prefix matching catches any in-window capture, and basenames were verified by hand because shared publisher hosts produce cross-jurisdiction matches. The standing rule is **none_found** over a guess.

A usable in-window pre-period source exists for 15 of 25 localities (60%): 11 archived code-page captures, 3 archived zoning-map PDFs, and San Francisco (in hand via the project’s own pipeline plus DataSF historical zoning layers). The remaining 10 are 9 `none_found` plus Burlingame, whose only source is an NZLUD 2019–21 proxy that is post-SB-35 and therefore not a valid pre-period at all. `confidence` reads high for exactly 15 rows and low for 10, and the CSV contains **zero live false-positive source URLs**. Rejected in the first pass: an Oro Valley, Arizona PDF on the shared `codepublishing.com` host that had polluted both Solano County and Walnut Creek; Palo Alto’s `Symposium.pdf`; and an SF SoMa area guide. Two more were killed in a second pass — **San Ramon**, whose 2015 `rezoning.pdf` is a seven-page “Rezoning to P-1 Submittal Requirements” *application*, not the citywide map (a lead survives: a `zoning_ordinance_map` page captured 2018, but it 302-redirects to EncodePlus and the underlying map’s vintage is unverified); and **Daly City**, whose 2014 “Zoning Ordinance.pdf” sits under a General-Plan-Update (`/gpu/`) path and could not be confirmed adopted (Wayback returned 404/503). Both were reset to `none_found`. Solano County (2015) and Walnut Creek (2014) survived as legitimate CodePublishing code-page captures; Palo Alto had already been reset.

The three surviving spatial anchors are the cleanest vintages — Sunnyvale 2014, Tiburon 2015 (a map effective 2006-03-31 still posted in-window, so the circa-2016 envelope was stable), San Leandro 2016 — while the code-page captures are 2014 (San Carlos, Walnut Creek), 2015 (Solano County), 2017 (Alameda, Mountain View, Oakland, Sonoma County) and 2018 (Clayton, El Cerrito, Hayward, Milpitas). That 2017–2018 cluster is comfortably before SB-9, SB-423 and the builder’s remedy but *after* SB-35 (2017), so for an SB-35-specific margin those are “early-ramp” rather than true pre-period and the 2014–2016 PDFs are the cleaner anchors. **The gap is patterned, not random:** the 9 `none_found` localities cluster by code publisher — eCode360 (Albany, Emeryville, Santa Rosa), American Legal *legacy* platform (Palo Alto, Fairfax, whose 2016 code sits on `library.amlegal.com` while the probe queried `codelibrary.amlegal.com`), public.law (Petaluma), a city-site PDF case (Piedmont), plus the two killed above — so the fix is a per-publisher amendment/superseded-version dig, a bounded build task. But the whole probe is downstream of the unmade envelope-definition decision: located sources are worth little until the extraction target is defined.

5 The treatment variable: builder’s-remedy exposure

`hcd_preemption_panel/hcd_preemption_panel.csv` (25×10) is the “preemption ramp” variable that dates the shock and supplies the instrument for the reaction-function estimation. The measure is exposure to California’s **builder’s remedy**: while a locality lacks a Housing Element found in substantial compliance by HCD it cannot deny qualifying affordable projects on zoning or general-plan grounds, so exposure runs from the statutory adoption deadline to the HCD compliance finding — rule-driven timing plausibly orthogonal to a locality’s contemporaneous demand shock.

The authoritative input is HCD’s *Housing Element Review and Compliance Report* (<https://data.ca.gov/dataset/housing-element-compliance-report>), pulled unmodified to `_source_data/hcd.csv` — 539 rows $\times$ 10 columns, one per California jurisdiction, last updated 2026-06-05 — filtered to `Record Type = Adopted`; prohousing status comes from HCD’s *Prohousing Designated Jurisdictions* list as of 2026-03-26. The sample is the 25 localities with a verified minutes start year of 2016 or earlier at high/medium confidence. Two date conventions are deliberately not collapsed: `he_adoption_date` (HCD’s *Date Received*, a proxy for self-adoption) and `hcd_certification_date` (HCD’s *Reviewed Date*, the formal finding). With the ABAG sixth-cycle deadline $T_0 = 2023-01-31$,

$$\text{exposure}_j = \max\{0, \text{months}(T_0 \rightarrow \text{HCD compliance finding}_j)\},$$

with $\max\{0, \text{months}(T_0 \rightarrow \text{HE received}_j)\}$ recorded alongside; localities certified before the deadline are floored at zero, not given negative exposure.

The variable is clean and the variation is strong. All 25 matched an HCD adopted-element record, all 25 carry `confidence = high` and sourced dates, and there are no `no_hcd_match` rows. Exposure ranges from **0.0 to 38.4 months** (mean 9.9, median 8.6): three localities have zero exposure, 22 positive, nine exceed 12 months and two exceed 24 months. 13 of 25 are prohousing-designated; five have sourced designation dates (Mountain View and Petaluma January 2024, Walnut Creek 2024-08-01, Santa Rosa and Sonoma County 2025) and the other eight carry `prohousing_date_to_verify`(HCD tracker XLS).

Table 1: Builder’s-remedy exposure, 25-locality estimation sample (deadline 2023-01-31 $\rightarrow$ HCD compliance finding). “designated” = prohousing-designated, exact date to verify; “—” = not designated.

Locality	HE received	HCD finding	Exposure (mo.)	Prohousing
Alameda	2022-11-16	2022-12-20	0.0	designated
San Francisco	2023-01-31	2023-02-01	0.0	designated
San Leandro	2022-12-09	2023-02-02	0.0	designated
Emeryville	2023-01-20	2023-02-03	0.1	designated
Oakland	2023-02-13	2023-02-17	0.5	designated
San Ramon	2023-02-06	2023-02-27	0.9	—
Milpitas	2023-05-16	2023-05-17	3.5	—
Petaluma	2023-03-22	2023-05-18	3.6	2024-01
Mountain View	2023-04-26	2023-05-26	3.8	2024-01
Hayward	2023-06-07	2023-07-27	5.9	designated
El Cerrito	2023-08-17	2023-08-22	6.7	designated
Albany	2023-08-16	2023-09-08	7.2	—
Tiburon	2023-09-22	2023-10-18	8.6	—
Walnut Creek	2023-08-25	2023-10-24	8.8	2024-08-01
Sonoma County	2023-08-30	2023-10-26	8.8	2025
Piedmont	2023-09-11	2023-11-09	9.3	—
Sunnyvale	2024-02-29	2024-03-06	13.2	designated
Burlingame	2024-03-15	2024-03-20	13.6	—
Fairfax	2024-03-14	2024-04-08	14.2	—
Solano County	2024-02-09	2024-04-09	14.3	—
San Carlos	2024-04-19	2024-04-25	14.8	—
Palo Alto	2024-07-26	2024-08-20	18.6	—
Daly City	2024-11-13	2024-12-03	22.1	—
Santa Rosa	2025-07-28	2025-09-12	31.4	2025
Clayton	2026-02-12	2026-04-13	38.4	—

The earlier 14-city panel is not a subset of this one. `zoning_envelope_project/hcd_preemption_exposure` holds 14 rows, all planning period `6S`, record type `Adopted`, current compliance `In`, anchored at **2023-02-01** rather than 2023-01-31, with exposure spanning **0.5 to 40.0 months**: Oakland 0.5 (found compliant 2023-02-17), Berkeley 0.9 (2023-02-28), Fremont 1.6 (2023-03-22), Redwood City 1.8 (2023-03-27), Mountain View 3.8 (2023-05-26), Richmond 8.0 (2023-10-02), South San Francisco 9.6 (2023-11-20), San Jose 11.9 (2024-01-29), Sunnyvale 13.1 (2024-03-06), Burlingame 13.6 (2024-03-20), Palo Alto 18.6 (2024-08-20), San Bruno 20.7 (2024-10-21), Daly City 22.1 (2024-12-03), San Mateo 40.0 (2026-06-01). Eight localities — Berkeley, Fremont, Redwood City, Richmond,

San Bruno, San Jose, San Mateo and South San Francisco — appear *only* in this 14-row file, the two files use different column schemas, and they anchor exposure differently, so neither should be deleted and the anchor convention must be unified before they are used together. San Mateo is the striking case: roughly 3.3 years of continuous exposure, certifying only in mid-2026 while its Peninsula neighbours cleared in months — an attractive treatment case, provided the minutes side can retrieve its hearings, which the minutes pilot could not even confidently classify.

Caveats. The **snapshot caveat** is the main analytical limitation: the HCD dataset is a *current-status* snapshot and all 25 read as “In” compliance today, so any decertification-then-recertification sequence during the ramp is invisible and exposure would be understated; **status_sequence** flags this on every row, and verifying contested cases is the one substantive **to_verify**. At the full ABAG level `hcd.csv` shows 108 In and one Out (Half Moon Bay, reviewed 2026-02-02). Also open: whether exposure ends at self-adoption or at HCD’s finding (the gap is material — Solano County 2024-02-09 vs. 2024-04-09); the eight unsourced prohousing dates; the 25-versus-36 sample width; and a possible *second* preemption series from HCD’s SB-35/SB-423 Statutory Determinations, RHNA-progress-based and not built here.

6 Data inventory

Bulk data lives on Dropbox; everything here is a small derived table, a script, or a narrative report. Status is *current* (believe it), *provenance* (kept for the record, do not cite), *superseded*, *input* (unmodified third-party), or *broken* (will not run as written; §7). Rows are grouped by the probe directory that produced them.

File	What it is	Status
bay_area_census/ — the frame		
<code>bay_area_locality_census.csv</code>	109 × 20; platform, URL, access, solution class, tier, confidence, both sides	current (the spine)
<code>raw/*.json</code> , <code>ca_place_geoid.json</code>	8 per-county surveys (108 localities); Gazetteer name → GEOID	provenance / current
<code>consolidate.py</code>	Builds the CSV, appends SF, attaches GEOIDs	current
<code>bay_area_locality_census_report.md</code>	Narrative roll-up, 2026-06-08	provenance; differs from CSV in 4 places
<code>akamai_civicplus_probe_finding</code>	The 28-host CivicPlus sweep table	current
<code>sample_saratoga_pc_minutes.2026-04-08.pdf</code>	19 KB; the one document retrieved end-to-end	evidence
archive_depth_probe/ — the time dimension		
<code>archive_depth.csv</code>	109 × 10; post-merge authority on minutes depth	current
<code>archive_depth_api.csv</code>	55 × 11; API depth-read detail	current
<code>consolidate.py</code>	Builds the CSV from <code>raw/</code> + census GEOIDs	current; “verified” filter is buggy
<code>api_probe.py</code>	Live API client (Legistar, CivicClerk, PrimeGov, Granicus RSS)	broken (/tmp/ I/O)
<code>assemble_api.py</code>	Merges 19 verified years in; results transcribed literally	current
<code>raw/*.json</code>	8 per-county shallow-probe observations	provenance
<code>archive_depth_report.md</code>	Shallow-probe narrative; carries a banner	superseded

File	What it is	Status
archive_depth_api_report.md	API-pass narrative; <i>no</i> banner	superseded (says 64/32, not 68/36)
civicplus_depth_probe/ — AgendaCenter depth		
civicplus_depth.csv	14 × 8; 2 resolved, 12 honest	unknown
final_pass.py	Single-year Search probes; supersedes probe.py	current
probe.py	First pass, wide 2004–2018 query	broken (hard-coded /Users/danpost/)
_results.json	Raw first-pass output	provenance; contradicts CSV twice
migration_cliffs_probe/ — the 2024 portal moves		
migration_cliffs.csv	3 × 8; legacy platform, reachability, legacy depth	current (no script)
minutes_platform_pilot/ — the 14-city pilot		
minutes_platform_pilot_report.md	Platform table, scrape results, transfer test, 13 review items	current on transfer; superseded on coverage
pilot_scrape.py	Daly City CivicClerk OData retrieval; stdlib only	current
jurisdiction_mappings.py	Proposed additive vocabulary layer; not wired in	current; path constant is depth-sensitive
sample_dalycity_pc-2023-12-05_minutes.txt	4.4KB; the document the transfer test rests on	evidence
zoning_map_form_probe/ — spatial form		
zoning_map_form.csv	109 × 8; form, URL, source type, download flag	current
consolidate.py, raw/*.json	Assembles the CSV; 8 per-county form probes	current / provenance
zoning_map_form_report.md	Narrative; <i>no</i> banner, numbers agree with the CSV	stale in status only
preperiod_envelope_probe/ — the 2016 envelope		
preperiod_envelope.csv	25 × 9; 15 usable, zero live false positives	current
wayback_probe.py, _wayback_results.json	Wayback CDX query 2014–2018; raw output, 25 records	current / provenance
finalize.py	Applies hand corrections and writes the CSV	broken (regenerates 17/25)
preperiod_envelope_report.md	Original write-up; carries a correction note	superseded on 17/25
hcd_preemption_panel/ — the treatment variable		
hcd_preemption_panel.csv	25 × 10; exposure 0.0–38.4 months	current, load-bearing
build_hcd_panel.py	Deterministic builder; prints the exposure summary	broken (/tmp/ sample list)
zoning_envelope_project/ — the 14-city assessment		
zoning_envelope_assessment.md	NZLUD coverage, 14-city code table, extraction pilot, panel, 15 review items	superseded on coverage; one claim contradicted
nzlud_14city_subset.csv	The 4 matching NZLUD rows × 19 columns	current
hcd_preemption_exposure_panel.csv	14 rows; exposure 0.5–40.0 mo., anchored 2023-02-01	current; <i>not</i> a subset of the 25-row panel
pilot_extract.py	Fremont/San Mateo code read; documents the Municode block	current (network-dependent)

File	What it is	Status
wrluri_crosscheck.py / .csv	14 × 8; NZLUD vs. WRLURI 2006/2018	CSV current; script broken
_source_data/ — unmodified third-party releases		
nzlud_muni.csv	Full NZLUD: 2,639 municipalities × 77 columns	input
hcd.csv	Full HCD compliance report: 539 rows × 10 columns, updated 2026-06-05	input

Several scripts reach sideways with `HERE.parent / "sibling"` (`archive_depth_probe/consolidate.py`, `archive_depth_probe/assemble_api.py` and `zoning_map_form_probe/consolidate.py` → `bay_area_census/`; `hcd_preemption_panel/build_hcd_panel.py` → `_source_data/`), so moving this group as a unit is safe and splitting it is not. `minutes_platform_pilot/jurisdiction_mappings.py` additionally counts `.parent` up to the repo root and must be re-counted on any change of depth — when that constant is wrong the import of the canonical enums fails silently and the validation degrades to a no-op.

7 Known contradictions and stale artifacts

The most actionable section in the sprint: places where two files in the repository disagree, or where a script would not reproduce the file it supposedly builds. None has been silently fixed.

7.1 Artifacts that contradict each other

1. **The archive-depth .md reports understate their own CSV.** `archive_depth_api_report.md` says known depth is **64 of 109** and ≤ 2016 breadth is **32**, whereas `archive_depth.csv` gives **68** and **36** after the CivicPlus and migration-cliff merges — and it carries *no* supersession banner. `archive_depth_report.md` is the older shallow snapshot (55 unknown, 27 verified, Marin County at 2024, gaps 2) and does carry one. Believe the CSV.
2. **consolidate.py’s “verified” filter wrongly rejects API-verified rows.** It defines the verified subset by rejecting any row whose flags contain `minutes_type.unverified` or `oldest_year.minutes.unverified`. The API merge added `api_verified` to those rows but never cleared the old shallow-probe flags, so the filter now rejects rows that were in fact document-verified — Alameda 2005, Oakland 2014, Hayward 2015 and Santa Rosa 2016 among them. Re-running it yields **38 of 109** “verified” rows, an artefact of stale flags rather than a real count. *Use the **confidence** column, not the flag filter.*
3. **civicplus_depth_probe/_results.json says Campbell 2018; the CSV says 2006.** The JSON records “21 Minutes files, years 2018–2018” from the wide 2004–2018 query. The CSV’s 2006 is the later and correct value — the wide query returned a truncated or most-recent slice, which is exactly why the second pass switched to single-year probes, and the year-by-year hits support 2006. The same JSON also records Los Altos Hills as **unknown** with `cid_not_found` where the CSV records 2022 from the static-HTML floor. Both stale values remain in the JSON.
4. **Millbrae’s flag contradicts its own method field.** Its `method` says “CID=8 found but Search returned no minutes 2008–2016” — the CID *was* discoverable and the window

came back empty — but its `flags` value is `civicengage_js_gated(CID_not_static)`, which describes the other eleven. Millbrae is a different failure mode and should be re-probed on its own terms, including whether `CID=8` is actually the planning body.

5. **The census report disagrees with the census CSV in four places.** Real discrepancies, not rounding; the CSV is authoritative:

Quantity	Report says		CSV says (authoritative)	
<code>civicplus_clean</code>	/	27 / 1		26 / 2
<code>civicplus_akamai</code>				
Overall tractability tier	<code>needs_adapter</code>	70,	<code>needs_adapter</code>	69,
	<code>access_blocked</code> 2		<code>access_blocked</code> 3	
<code>minutes_access</code> (report §B)	<code>clean</code> 66, <code>unknown</code> 23,		<code>clean</code> 78, <code>unknown</code> 10,	
	<code>akamai_403</code> 2		<code>akamai_403</code> 3	
<code>in_nz_lud</code> filled	8 pilot cities, 4 yes / 4 no		11 rows filled, 4 yes / 7 no	

The third row is the report contradicting *itself*: its own methods caveat already carries the post-sweep numbers while its summary §B was never updated. Its §A table sums the CivicPlus class to 28 either way.

6. **The zoning assessment’s “4 of 109 met the deadline” is not reproducible from `hcd.csv`.** `zoning_envelope_assessment.md` writes that only four Bay Area jurisdictions met the 2023-01-31 deadline — Alameda, Emeryville, San Francisco, San Leandro — and that *none of the 14* did. Filtering `hcd.csv` to the 109 ABAG rows gives neither reading. **Seven** jurisdictions have a `Date Received` strictly before the deadline (Alameda 2022-11-16, San Leandro 2022-12-09, Sebastopol 2023-01-09, Emeryville 2023-01-20, Fremont 2023-01-23, Berkeley 2023-01-24, Rohnert Park 2023-01-26), with San Francisco landing exactly on 2023-01-31; and only **one** — Alameda — has an `HCD Reviewed Date` before the deadline (2022-12-20). Two of the seven, Berkeley and Fremont, are in the 14-city set, so the claim that none of the 14 adopted in time is contradicted by the data file. The substantive point survives either way, since HCD’s *finding* is what ends exposure and only Alameda had one before the deadline — but the sentence should be corrected.
7. **The two HCD panels disagree by construction.** The 25-row panel anchors at 2023-01-31 and spans 0.0–38.4 months; the 14-row panel anchors at 2023-02-01 and spans 0.5–40.0 months, with eight localities appearing only in the smaller file (§5).

7.2 Scripts that will not reproduce their own output

8. `build_hcd_panel.py` reads a `/tmp/` file that no longer exists. It takes its sample definition from `/tmp/preperiod_sample.json`, a path outside the repo, so the script is not re-runnable end-to-end. The 25-locality list should be committed alongside it before the panel is rebuilt.
9. `preperiod_envelope_probe/finalize.py` never encoded the 17→15 correction. The San Ramon and Daly City kills were applied to `preperiod_envelope.csv` *in place*, not through the script: `finalize.py` still sets San Ramon to `confidence = med` and Daly City to a valid text source, so re-running it as written would regenerate the stale 17/25 file. Fold the two corrections into its `CORR` table first. (It also reads `/tmp/preperiod_sample.json`.) The

original report’s headline “17/25 (68%)” is superseded; a one-line correction note pointing to 15/25 sits at its top, and the rest of its method and platform diagnosis still stands.

10. **Three more hard-coded paths.** `archive_depth_probe/api_probe.py` writes to `/tmp/api_probe_results` and `assemble_api.py` reads `/tmp/api_targets.json`, neither in the repo — though the verified results are transcribed literally into `assemble_api.py`, so the merge itself is auditable. Both CivicPlus scripts read their 14-locality target list from `/tmp/cp14.json`, and `probe.py` additionally writes to a `/Users/danpost/...` path that does not exist on this machine, so it is broken as written (`final_pass.py`, writing to `/tmp/cp14_final.json`, is fine). `wrluri_crosscheck.py` hard-codes the same defunct `/Users/danpost/...`. Dropbox root and needs the data root taken from the environment.
11. **Two reports are stale in status rather than content.** `zoning_map_form_report.md` carries no supersession banner, but its numbers — 67/28/13/1, the 55-versus-12 split, 71 apparent downloads — all re-derive from the CSV. `minutes_platform_pilot_report.md`’s platform calls are superseded by the census wherever they differ, and its Akamai framing is narrower than what was later measured: the 28-host sweep found only two blocked CivicPlus sites region-wide, so Fremont is an exception rather than evidence about the platform. Fremont’s own 403 is confirmed in both. The two consolidating documents these probes originally reported into — `output/DATA_STATUS.md` and `output/final_recon_bundle_report.md` — were deleted on 2026-08-30; their content is folded into this memo.

7.3 What is unverified across the board

No minutes document was retrieved anywhere in the recon except the two evidence files (the Saratoga PDF and the Daly City text). Every depth year in `archive_depth.csv` rests on a listing or an API metadata record — the 54 low-confidence rows and the 30 `type_ambiguous` rows have never been checked against an actual document, and the migration-cliff years likewise rest on feed entries. No zoning layer in `zoning_map_form.csv` has been downloaded, opened, or joined to a parcel; `download_apparent` records that a path *looked* available, not that it works. Four localities may have no written minutes online at all — Vallejo (agendas and video to 2006, no minutes-type files), Sebastopol (video only), Benicia (council minutes not located) and Dixon (1998–2009 on DVD at City Hall) — candidates for the deferred physical-records workstream. The remaining access-blocked or unreachable sites are Fremont, Portola Valley and San Jose (Akamai), Redwood City and South San Francisco (unsupported platform), Fairfield and Suisun (403), and American Canyon and Calistoga (refused connections, possibly transient). Also unverified: San Jose’s by-right read (blocked by Municode; the city posts a static Title 20 PDF that would resolve it), the QCode paths for South San Francisco and Burlingame, every NZLUD height field for a city whose code defers height to a General Plan map, all overlay districts, Berkeley’s and the City of San Mateo’s minutes platforms, Cupertino’s true minutes source, the city of Sonoma’s exact pre-2017 year, and whether Daly City genuinely seldom posts minutes or merely posts them elsewhere.